Consequa-X: An Evidence and Intervention Protocol for Consequential Agent Actions

Devrajsinh Jhala

***Abstract*—A lost acknowledgement can turn a successful agent action into a duplicate when the agent retries. Reliable diagnosis must distinguish what the agent reported, whether the requested state holds and which actor produced it. Consequa-X contributes an evidence protocol that preserves those distinctions, observation-specific provenance and explicit unknown verdicts. Its optional research adapter adds a declared on-reach intervention game and refusal-aware accounting around existing contrastive and permutation Shapley estimators. Across 2,048 scripted-simulator sampling trials, all 655,360 planned replay branches were retained, including invalid and unstarted branches. At one predeclared retry coefficient, nominal normal intervals covered the reference in 118/128 trials; Hoeffding bounds covered in 128/128 but detected its sign in only 8/128. A separate 24-episode, single-model exploratory validation found six contradicted status claims among ten assessable claims for one prompt. These results support testable protocol properties and expose the availability and precision costs of the declared evidence and replay rules. Comparative accuracy, real-service effectiveness and broad model generalization remain open.**

***Index Terms*—agent systems, outcome verification, evidence provenance, counterfactual replay, software artifacts, uncertainty**

I. Introduction

A consequential tool operation has distinct stages: selection, dispatch, execution, response, reported assessment and external observation. A service can commit an effect before its acknowledgement is lost. Retrying can create another effect unless the logical operation's identity and idempotency semantics are retained. Conversely, a successful response can coexist with an incomplete or incorrect effect. Explaining such runs requires more than reading the last tool response or the agent's prose.

State satisfaction and effect origin also differ. An external refund may satisfy the requested balance or count while remaining unlinked to this agent. A source-supported origin link is a provenance claim; it does not establish counterfactual necessity. A replay effect is a separate claim about a declared intervention, continuation policy and outcome distribution.

Consequa-X makes these distinctions inspectable. Its core contains typed records, schema validation, application verification and evidence-only explanation. Its research adapters provide isolated refund worlds, public-only policy inputs, a local Ollama adapter and bounded replay experiments. Reported assessments are captured outputs, not measurements of hidden reasoning.

The paper centers three testable protocol claims:

1) **C1: State and origin remain separate.** A satisfied contract can retain an other-actor or unknown origin. Successive readbacks retain their own supporting witnesses rather than transferring provenance between observations. Verification and graph tests exercise external-actor, missing-provenance and conflicting-witness cases; the local validation contains an other-actor example.
2) **C2: The replay game is explicit.** On-reach semantics define how normal early termination treats later requested holds. Strict and on-reach results remain separate. Analytic enumeration and the variable-length controls test this distinction; Appendix A states its formal scope.
3) **C3: Refusal remains observable.** Every planned branch is retained with call accounting; incomplete evidence or an invalid draw refuses affected estimates. The repeated-sampling study accounts for all 655,360 slots and reports availability beside numerical accuracy.

These are implementation and declared-game claims. The contribution combines them in a portable protocol around existing estimators; an equally informed baseline comparison has not yet established their empirical advantage. Figure 1 separates the captured-evidence core from the explicitly invoked research replay path.

II. Related Work

Causal Agent Replay (CAR) performs same-policy interventions on recorded decisions, evaluates user-specified outcomes and uses contrastive suffix and permutation Shapley attribution [1]. Consequa-X implements an independent binary adaptation with its own typed-world, evidence and refusal boundaries. Applying a verified outcome to replay is therefore not an estimator novelty claim. Policy-Conditioned Counterfactual Credit studies changes in final verified success under specified interventions and a continuation policy [2]; its learning objective differs from post-hoc external-effect diagnosis, but verified targets and valid interventions are shared concerns.

CausalLoss-Fin jointly intervenes on agent decisions and repairable infrastructure messages [3]. Its deterministic programmatic evaluation limits transfer to stochastic language policies, but it already occupies the broad combination of financial effects and joint decision/fault attribution. A future comparison must expose applicable variables to every method rather than charge omitted infrastructure effects to decisions.

Verified Tool Calls distinguishes acknowledgements from independently checked effects and uses unknown postcondition conclusions [4]. Callability Is Not Operability studies controlled interface ambiguity and terminal agent claims [5]. Three-valued verification and response/effect separation are

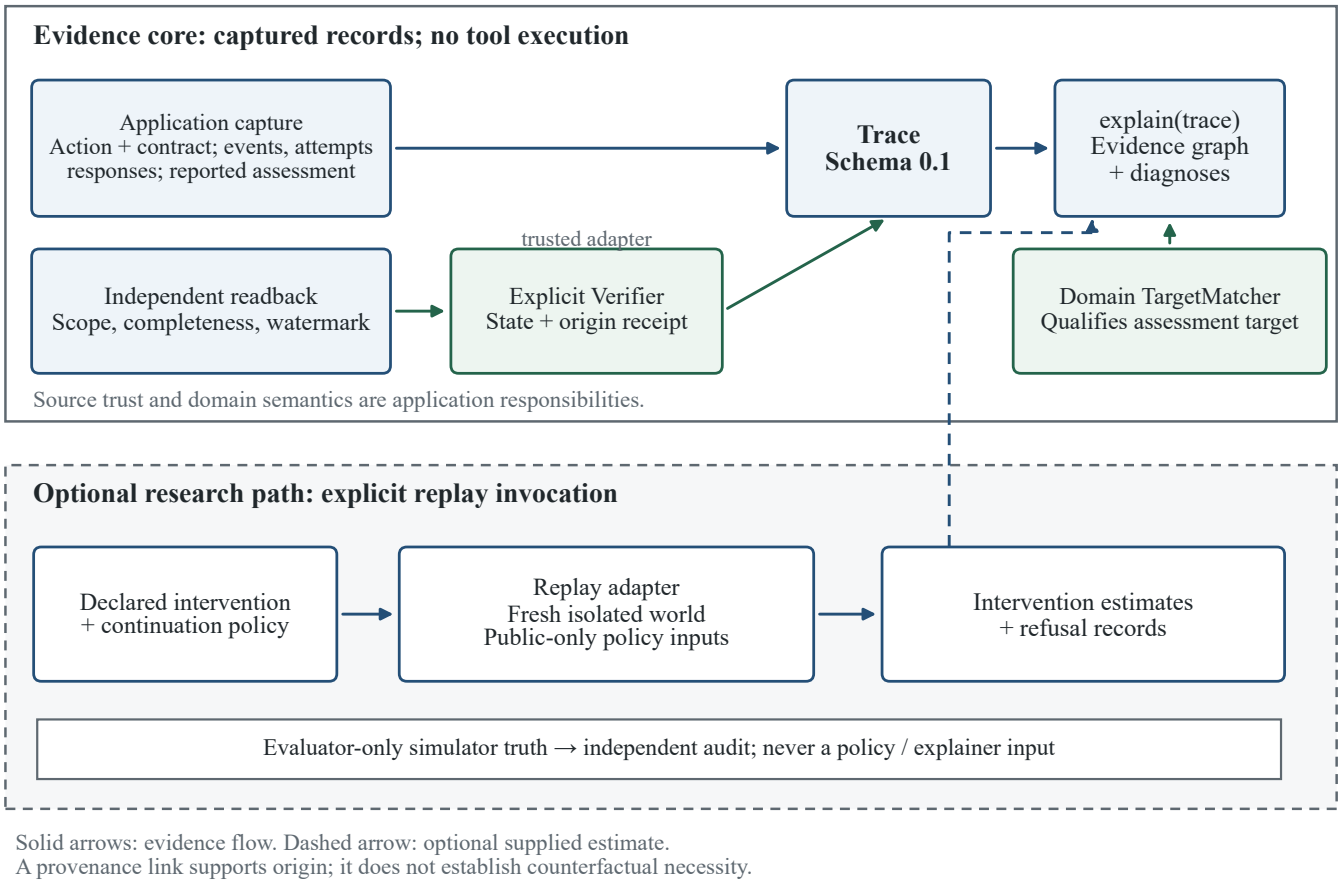


Fig. 1. Evidence flow and trust boundaries. Application verifiers interpret independent readback; explanations use captured evidence. Explicitly invoked replay can supply estimates, while evaluator truth remains audit-only. Source trust and domain semantics remain application responsibilities.

foundations and comparison requirements here. These targeted primary-source checks do not constitute a systematic review or an independent reproduction of those papers' empirical results.

Who&When attributes failed multi-agent runs to an agent and decisive step from annotated logs [6]; this differs from verifying a particular external effect from an independent witness. PROV-DM already represents entities, activities, agents and responsibility [7]. Consequa-X is neither a new general provenance model nor a PROV conformance implementation. Its narrower contribution is an executable protocol linking temporal contracts, observation witnesses, separate origin verdicts and explicit refusal.

Agent–tool anomaly work already separates external events from runtime observations, including exogenous effects [8]; LIMBO grades committed-effect histories and analyzes late commits [9]. Response/effect separation and effect ledgers are therefore prior concepts. Our study audits a particular protocol rather than establishing a safer recovery controller. LIMBO's late-commit argument assumes a keyless, non-idempotent write with no known finite in-flight bound and finite task-completion time. Under those conditions, bounded watermark polling cannot guarantee exactly-once execution. Idempotence and eventual consistency are established systems concerns [10], [11].

AppWorld, ToolSandbox and τ-bench already evaluate world-state outcomes or stateful trajectories [12], [13], [14]. They provide future transfer targets; this paper evaluates none of their task populations. For statistics, known binomial Wald-interval limitations [15] motivate careful interval checks but do not by themselves establish paired-Shapley undercoverage. Modern bounded-mean betting intervals [16] are planned comparators, with the sampling unit and stopping rule still requiring explicit validation for this replay setting.

III. Evidence Protocol and Core Interfaces

A. Objects, contracts and trust

Let H be visible history, A a logical action, E an execution attempt, R its response, B a timestamped reported assessment and O an independent readback observation. A contract C declares payload, count, deadline and stability horizon. The verifier conclusion $V(C, O)$ is satisfied, violated or unknown. Origin is separately confirmed, other actor or unknown. Evaluator-only simulator truth never becomes a policy or explainer input.

The built-in refund predicate uses integer currency minor units. An eligible effect matches payment, amount and currency, commits in the declared time window and persists through the horizon. Exact-count contracts also constrain the count during the stability interval. A stale watermark or inadequate scope and completeness yields unknown rather than a confident absence claim. Source authority, scope, completeness and watermark declarations remain trusted adapter inputs. Schemas and hashes validate structure and changed bytes, not source authenticity or real transactional guarantees. Figure 2 illustrates why a response and a reported assessment cannot substitute for a horizon-covering outcome observation.

B. Reusable application boundary

The evidence core can operate on a production application's captured records without any replay. A read-only adapter obtains independent observations; verification returns state and origin conclusions, and explanation renders the supporting evidence. This path requires neither simulator reset nor a second execution of a consequential action.

Applications can supply an explicit read-only verifier through `verify(action, observation, verifier=...)`. The wrapper checks receipt type, contract identity, observation identity, source, watermark and timing. Domain predicate semantics, authentication and read-only adapter behavior are the application's responsibility. There is no evaluation of arbitrary predicate strings. An offline artifact-delivery example constructs captured records and checks independently readback content digests without importing the refund simulator or executing writes.

`explain(trace)` reads a captured verification and performs no replay. Refund target matching is built in; other predicates receive a neutral explanation unless a trusted domain matcher explicitly qualifies attributable effects against payload and temporal rules. Matchers can return only unique IDs within the receipt's attributable IDs. Unknown receipts are not interpreted as target matches. A qualified provenance link still is not a causal effect.

C. Successive witnesses and graph meanings

Repeated readbacks can disagree about the same effect identifier. The graph therefore retains observation-specific witnesses and an identity-only shared effect node. Dispatch-to-effect edges attach to the witness supplying origin metadata; verification edges use witnesses in that receipt's own observations. A later conflicting actor, corrected payload or reversal cannot overwrite an earlier witness and inherit its unqualified origin link. Shared identifiers do not authenticate or reconcile the records. Estimated-influence edges require captured event identifiers and an explicitly supplied intervention estimate.

IV. Replay Targets, Games and Uncertainty

A. Targets and conditional suffix effects

The refund adapter exposes three targets: initial action selection, global state-contract satisfaction and action-linked consequence. A known global violation is false for the linked target. Satisfied state with exclusively external qualifying effects is false for that target; insufficient provenance can instead yield unknown. The linked target combines global satisfaction and qualifying operation-linked witnesses rather than reconstructing an agent-only world. Unknown observations do not acquire numeric labels.

For known binary target Z, set bad outcome $X = 1 - Z$. Prefix replay holds factual decisions before position k and resamples the suffix. With one observed factual bad label x_f, the conditional estimand is

$$\gamma_k = \mathbb{E}[X \mid \text{prefix held, suffix resampled at } k] - x_f. \quad (1)$$

The factual term is one observed constant, not a population rate estimate. This factual-anchored target answers: how often would a fresh suffix avoid the bad outcome observed in this episode? It is useful for episode diagnosis, but does not estimate the effect of changing one decision against an independently resampled original arm. That population contrast would instead compare two expectations, for example $\mathbb{E}[X \mid \text{hold } k] - \mathbb{E}[X \mid \text{resample } k]$, with the other conditions declared. Neither estimand is generally preferable; they answer different questions and have different sampling costs. Here negative γ_k indicates rescue relative to the observed episode. An earlier irrelevant position can have a nonzero suffix effect because its suffix contains a later stochastic choice. The *pivot* is the latest position with negative population suffix effect under this definition. The retained nominal selector attempts to detect that position only if all planned positions are estimable. Its pointwise intervals give no simultaneous or post-selection coverage guarantee; selected pivots are exploratory diagnostics rather than unique causes.

B. Held-decision coalition games

Let N be reached factual positions. Coalition S requests holding captured decisions at positions in S; other decisions are sampled. Reached held decisions must match the typed public stage. In the *strict* game, an earlier termination leaving a requested held position unreachable invalidates the draw. Finite samples avoiding invalidity do not define the population game.

The separate *on-reach* game treats requested positions beyond a normal termination as inactive and uses the actual completed episode. Reached incompatible operations remain invalid; provider failures and exhausted budgets are not normal inactivity. This is an explicit alternative game, not a repair of retained strict results.

For a defined game $v(S) = \mathbb{E}[X \mid S]$, signed Shapley values allocate the held-decision effect [17]:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[v(S \cup \{i\}) - v(S) \right]. \quad (2)$$

Fresh forward/reverse permutation walks form one paired contribution vector. Independent empty/full draws diagnose the endpoint scale. Values are signed allocations under the declared game, not normalized responsibility shares. Positive values mean holding the factual decision increases bad outcomes; negative values mean it is protective.

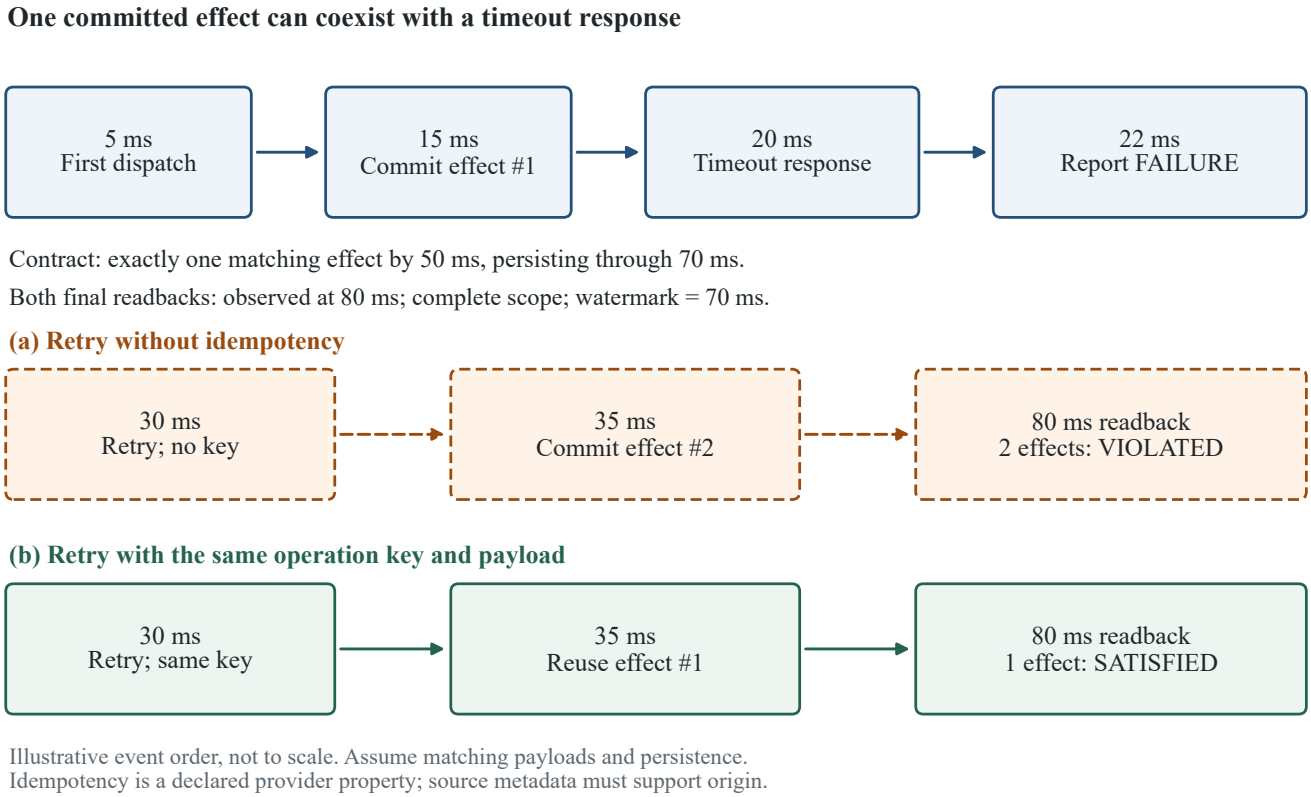


Fig. 2. Illustrative timeout-after-commit sequence. A committed effect precedes a timeout and reported failure. An unsafe retry creates a duplicate; reusing the same provider-scoped idempotency key and payload preserves one effect. Complete final readback covers the stability horizon. Arrows show event order, and times are illustrative rather than sampled observations.

C. Intervals and refusals

Contrastive defaults use exact binomial Clopper–Pearson bounds translated by $-x_f$ [18]. Signed pair observations lie in $[-1, 1]$; the default Hoeffding half-width for P independent pairs and failure allowance α is [19]

$$h = \sqrt{\frac{2\log(2/\alpha)}{P}}. \tag{3}$$

This is the half-width before clipping the interval to $[-1, 1]$. Endpoint differences use two exact binomial marginals with a split failure allowance. Each marginal has confidence $1 - \alpha/2$; the difference interval is $[L_f - U_e, U_f - L_e]$. The union bound requires valid marginal bounds, not independence between the two arms. Bootstrap and normal intervals remain explicit legacy comparators. These are standard conditional methods with sampling assumptions and precision costs, not distribution-free guarantees for dependent model continuations. All reported coefficient intervals are pointwise. For a predeclared family of m valid marginal intervals, allocating failure probability α/m to each would give a Bonferroni family guarantee without independence across positions. That adjustment was not used in these retained studies. A nominal latest-rescue selection therefore does not inherit a 95% family guarantee. Variance-adaptive or betting-based bounded-mean methods [16] are relevant future comparators; the present results do not establish that Hoeffding is optimal or that conservative uncertainty must have this width.

Replay requires a resettable, isolated simulator or a validated digital twin. It must not reissue real refunds or other consequential production writes. The studies below reset a scripted payment simulator, whose fidelity is a separate assumption; the evidence-only production path does not require it. Held replies consume no fresh calls; every attempted fresh call, including a provider/protocol failure, consumes budget. Every planned branch remains recorded. Unknown, invalid, failed and unstarted branches are not dropped or redrawn; affected estimates are refused. Targets and interval variants reuse captured evidence without extra policy calls.

D. Analytic refund reference

Consider the declared on-reach refund mechanism with adequate budget, complete horizon-covering readback and, for the linked target, sufficient provenance. The planted factual decisions are selection s, retry r and final stop t. Fresh selection occurs with probability p and fresh timeout retry with probability q under the declared independent choice law. The first attempt commits despite its timeout; without idempotency, a retry commits a second refund. Exactly one effect therefore occurs with probability $p(1 - q)$. For the consequence bad-

TABLE I

ORIGINAL PROTECTIVE-STOP CONSEQUENCE POSITION: NOMINAL 95% PERCENTILE-BOOTSTRAP CONTRASTIVE AND NORMAL PAIRED-SHAPLEY COVERAGE. EACH ENTRY HAS 128 AVAILABLE, REFERENCE-ELIGIBLE TRIALS.

Retry q	Contrastive covered	Shapley covered
0.1	100/128	107/128
0.5	122/128	121/128
0.9	103/128	103/128

TABLE II

FRESH PROTECTIVE-STOP SHARED-DRAW COVERAGE: 32 TRIALS PER ROW. BOOTSTRAP/EXACT-BINOMIAL INTERVALS COVER THE CONTRASTIVE EFFECT; NORMAL/HOEFFDING INTERVALS COVER THE PAIRED-SHAPLEY COEFFICIENT.

q	Bootstrap	Exact binomial	Normal	Hoeffding
0.1	26/32	31/32	23/32	32/32
0.5	31/32	32/32	32/32	32/32
0.9	31/32	32/32	29/32	32/32

outcome game, the four relevant coalition values are

$$\begin{aligned} v(\varnothing) &= 1 - p(1-q), & v(\{s\}) &= q, \\ v(\{r\}) &= 1, & v(\{s,r\}) &= 1. \end{aligned} \tag{4}$$

The final stop is a dummy: adding t changes none of these values. Averaging the two orders of the active players gives

$$\begin{aligned} \phi_s &= -\tfrac{1}{2}(1-p)(1-q), \\ \phi_r &= \tfrac{1}{2}(1+p)(1-q), \qquad \phi_t = 0. \end{aligned} \tag{5}$$

Their sum is the endpoint difference $p(1-q)$; the selection/retry interaction $v(\{s,r\}) - v(\{s\}) - v(\{r\}) + v(\varnothing)$ is $(1-p)(1-q)$. At $p = q = 0.5$, these are shares $[-0.125, 0.375, 0]$, endpoint 0.25 and interaction 0.25. With idempotency, any selected refund creates one effect, so consequence shares become $[-(1-p), 0, 0]$, as for initial action selection; the interaction is zero. These are population references derived from the declared mechanism, not fitted to sampled outcomes. A call cap can refuse the finite estimate despite a defined ideal reference. Unknown evidence or strict unreachable-held decisions do not acquire numeric references from these forms.

A standalone standard-library check uses exact rational arithmetic on a five-by-five (p, q) grid, two idempotency settings and two targets: 100 control cases. It checks all coalition values and six permutation orders against these forms without importing the framework or reading sampled study data. This checks the stated mathematical law, not simulator execution or sampling validity.

V. AUDITED EVALUATION

A. Original calibration and finite analysis

The *protective-stop* fixture forces the factual timeout decision to stop after one committed refund. Fresh continuations retry with probability q, so resampling can introduce a duplicate. The *retry-pivot* fixture instead forces a factual retry; its timeout position is the latest population rescue position when $0 < q < 1$. These are planted scripted decisions, not model-derived causal labels.

The original development study retained 1,280 trials across ten specified mechanisms, with 128 repetitions. Of 223,232 planned branches, 198,690 completed, 4,062 were invalid, 256 failed at the deliberate cap and 20,224 were unstarted. Standalone audit scripts checked raw evidence, seeded outcomes, statistics and accounting. Table I reports a preselected protective-stop consequence position with 128 available, reference-eligible estimates in each row.

Separate finite-distribution calculations yielded protective-stop normal pair coverage of 83.7775% at $q = 0.1, 0.9$ and 92.6377% at $q = 0.5$ for 16 pairs. For interior Bernoulli q, homogeneous draws force an empirical percentile bootstrap interval to a point, giving coverage ceiling $1 - q^K - (1-q)^K$. At $K = 16$ and $q = 0.1, 0.9$, this is about 81.47%. These findings concern the specified adaptation; they do not establish coverage of upstream CAR implementations.

B. Shared-draw conservative comparison

A separately frozen follow-up retained 384 trials across twelve cases, 66,048 planned branches and 87,054 actual scripted calls. Variants reused the same points, draws, seeds and refusals. Exact probability weights over sample counts 8,16,32,64 and eleven probabilities yielded minimum grid coverage 7.72% for the limiting empirical percentile bootstrap, 85.15% for Wilson and 96.68% for Clopper–Pearson. The bootstrap grid is its infinite-resample limit, distinct from the replay's finite seeded bootstrap.

Table II is the protective-stop timeout position, not every case. The full summary includes exact-binomial coverage 30/32 for a retry-pivot case. Finite observed coverage can fall below nominal despite a conservative population method. At $q = 0.1$, mean normal versus Hoeffding width was about 0.180 versus 1.358; at $q = 0.5$, 0.344 versus 1.184. Hoeffding detected the negative reference sign in 0/32 trials at both weaker settings, and 32/32 at $q = 0.9$. Wider intervals therefore impose a substantial detection cost.

The standalone saved-evidence audit checked its recorded derived-artifact inventory and rejected twenty targeted corruptions. A separate SciPy numerical audit covered 1,290 binomial cases and 1,290 endpoint differences, with maximum binomial endpoint disagreement approximately 2.66×10^{-14}. SciPy was an audit dependency, not a framework runtime dependency. Finite grids and synthetic replays do not establish live-model sampling independence or simultaneous attribution coverage.

C. Scripted-simulator interactions and variable length

An independent scripted-simulator enumeration covered four deterministic fresh-choice components, eight held masks, two games and eight cases, retaining 512 planned branches and 718 actual scripted calls. Outcomes came from ordinary payment readback rather than injected payoff values. With selection and timeout retry probabilities $p = q = 0.5$, unsafe retry and exactly-one verification, on-reach consequence

shares were $[-0.125, 0.375, 0]$, with selection/retry interaction 0.25. Action-selection shares were $[-0.5, 0, 0]$. Idempotency changed consequence shares to $[-0.5, 0, 0]$ and removed the interaction. This is a declared programmatic mixture, not observed model interaction recovery.

The strict population game remained undefined at interior probabilities; missing provenance, lag and small budgets retained their distinct limits. The independent audit rejected twenty-nine semantic corruptions. The original export manifest and data remain preserved alongside the derived canonical inventory. Two sampler integration controls exercised 32 pairs and 32 endpoint draws, with eighteen coefficients within a broad 0.3 regression tolerance. That tolerance alone is not a calibration result.

D. Repeated scripted-simulator sampling

A separately frozen protocol exercised the actual permutation sampler across all eight declared scripted-simulator mechanisms, both games and 128 fresh repetitions per mechanism/game: 2,048 trials. Each trial planted the same three factual decisions and used 32 forward/reverse permutation pairs, one draw per coalition evaluation and 32 independent draws per endpoint, yielding 320 planned branches. The planted factual path was specified rather than selected by seed search. The cap was 1,000 actual calls including three factual calls, except for the four-call budget control. Collection used normal intervals; Hoeffding and exact-binomial endpoint differences were rescored from identical draws without additional calls, altered points or redrawn refusals. The installed checkpoint, precollection protocol, all compressed trial/branch rows and replicate-zero full evidence with separately stored evaluator truth are retained.

The run retained all 655,360 branches: 536,150 completed, 37,423 invalid, 256 failed and 81,531 unstarted. It consumed 628,535 scripted calls, including 6,144 factual calls, and zero language-model calls. On-reach execution recorded 37,392 branches with inactive requested holds; these are normal-termination records, distinct from strict invalidity and budget exhaustion. Figure 3 retains both games and every branch status, displaying inactive holds separately from the mutually exclusive status partition.

A standalone auditor importing no framework implementation checked the retained trials, branches, 4,210 complete traces and all summary rows. Its receipt binds the frozen source revision and original 52-file manifest. Derived tables retain every row separately from that sealed collection; auditing made no generation calls.

Actual-call cost also depends on availability. The on-reach unsafe $(0.5, 0.5)$ stratum used 42,526 scripted calls for 128 available global-consequence coefficient vectors, or 332.23 calls per vector, including factual capture and independent endpoints. Each vector contains three jointly estimated players. The lagged stratum spent 42,424 calls for zero such vectors; its cost-per-usable vector ratio is undefined. Shared targets and interval variants do not multiply the number of independent usable vectors. These are scripted-call costs, not production latency or financial charges.

Table III reports every declared mechanism for one specified target and position, including unavailable controls. Figure 4 places coverage beside interval width and detection to expose their tradeoff without pooling targets or cases. At $(p, q) = (0.5, 0.5)$, the shared retry estimate had bias -0.00269 and RMSE 0.0611. Normal coverage 118/128 has a 95% Wilson interval $[0.862, 0.957]$; 128/128 has $[0.971, 1]$, not proof of perfect population coverage. The conservative intervals sharply reduced weak-effect sign detection. For the consequence stop dummy, normal false nonzero flags ranged from 0 to 14/128 across the six assessable mechanisms; Hoeffding flags were 0/128 in each. The deterministic zero-reference case yielded 128 normal point intervals; their correctness in that case does not validate point intervals for stochastic outcomes. With 128 repetitions, $[0.862, 0.957]$ also shows why an observed 92.2% rate does not resolve a small departure from 95% coverage. These runs are diagnostic integration evidence, not a precise empirical ranking of near-nominal methods. The finite-distribution calculation supplies a separate analytic check only for its declared independent Bernoulli law.

Endpoint behavior was separately assessed. For the on-reach unsafe $(0.2, 0.8)$ global-consequence endpoint, normal coverage was 92/128 (Wilson $[0.635, 0.789]$), with 35 point intervals and mean width 0.114. Exact-marginal differences covered 128/128, with mean width 0.319. These finite results demonstrate the declared precision tradeoff; they do not establish simultaneous coverage or calibration under live-model dependence.

In the on-reach game, every lagged-consequence coefficient was refused as unknown; every budget coefficient was refused after unstarted slots, despite the latter's defined ideal reference. Removing provenance preserved global-consequence availability but refused all 128 linked-consequence coefficient vectors. All nondeterministic strict coefficient vectors were unavailable and their population games undefined; the fully deterministic strict control remained defined. One finite strict interaction point at $(0.8, 0.2)$ did not make that population interaction defined. One strict linked-consequence endpoint without provenance was numeric but reference-ineligible, giving coverage 0/0 rather than an accuracy result. Pooled-coalition interaction diagnostics have no confidence interval and supply no coverage or sign-detection result. The full summary retains separate availability, reference eligibility, bias/RMSE, coverage uncertainty, width, flags, endpoints, interactions and refusal rows for every target, position, game and interval variant. Reused targets and variants are not independent replications.

E. Exploratory local-model validation

A frozen installed Qwen3 4B checkpoint completed six validation cases, two repetitions and two unchanged prompt profiles, using 59 actual calls within a 72-call cap. No episode failed or remained unstarted. Model identity, prompt/payload hashes, public traces and the exact source checkpoint are retained. The original and revised profiles induced different

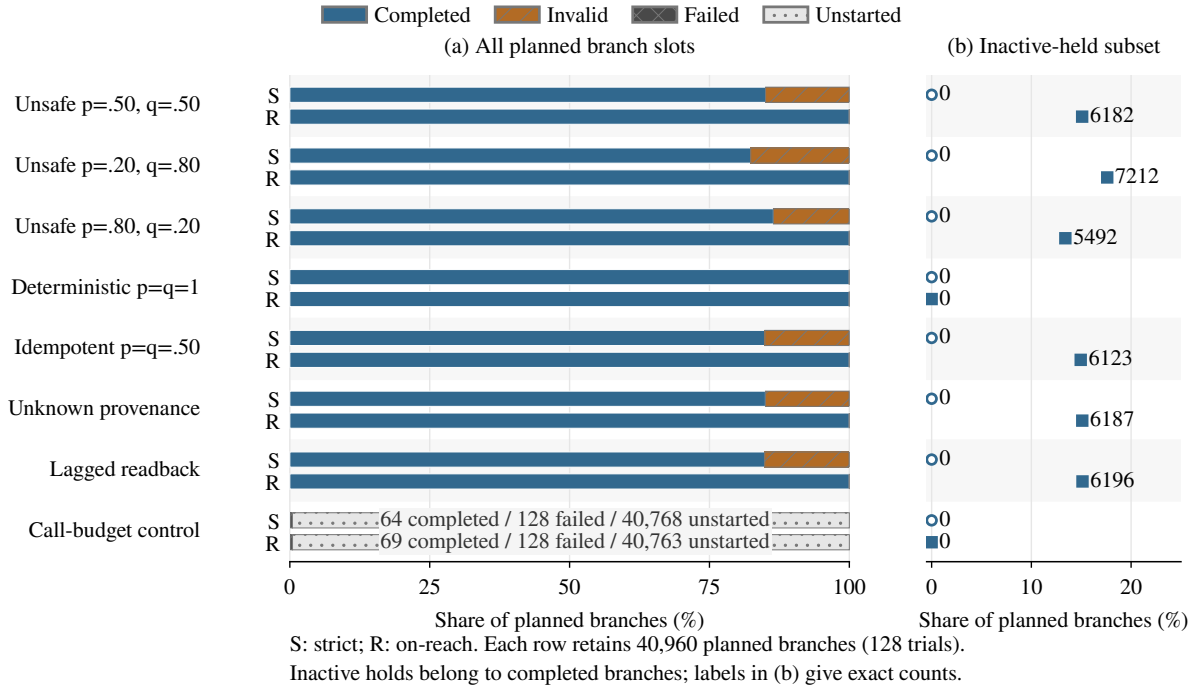


Fig. 3. Disposition of all planned branches across eight mechanisms and two games: 40,960 slots per row. Completed execution includes unknown target evidence and known violations; it does not establish an available coefficient. Failed and unstarted budget-control slots remain visible through exact counts. Inactive holds are a subset of completed branches, shown separately.

TABLE III

ALL EIGHT DECLARED CASES: ON-REACH RETRY COEFFICIENT FOR GLOBAL CONSEQUENCE, 128 PLANNED REPETITIONS PER ROW. N/H DENOTES NORMAL/HOEFFDING. AVAILABILITY A AND POPULATION-REFERENCE ELIGIBILITY R EACH HAVE DENOMINATOR 128. FLAGS DENOTE CORRECT DIRECTIONAL DETECTION FOR NONZERO REFERENCES AND FALSE NONZERO FLAGS FOR ZERO REFERENCES. DASHES DENOTE UNDEFINED QUANTITIES; 0/0 DENOTES AN UNDEFINED COVERAGE RATE. HOEFFDING INTERVALS ARE CLIPPED TO $[-1, 1]$, EXPLAINING THE 0.773 MEAN WIDTH AT $(0.8, 0.2)$ VERSUS THE UNCLIPPED 0.960 WIDTH AT THE TWO OTHER STOCHASTIC UNSAFE SETTINGS.

Case (p, q) or control	Reference	A ; R	Covered N / H	Mean width N / H	Flags N / H
Unsafe $(0.5, 0.5)$	0.375	128 ; 128	118/128 ; 128/128	0.226 / 0.960	128/128 ; 8/128
Unsafe $(0.2, 0.8)$	0.120	128 ; 128	124/128 ; 128/128	0.154 / 0.960	123/128 ; 0/128
Unsafe $(0.8, 0.2)$	0.720	128 ; 128	123/128 ; 128/128	0.218 / 0.773	128/128 ; 128/128
Deterministic $(1, 1)$	0	128 ; 128	128/128 ; 128/128	0 / 0.960	0/128 ; 0/128
Idempotent $(0.5, 0.5)$	0	128 ; 128	123/128 ; 128/128	0.244 / 0.960	5/128 ; 0/128
No provenance $(0.5, 0.5)$	0.375	128 ; 128	120/128 ; 128/128	0.228 / 0.960	128/128 ; 7/128
Lagged readback $(0.5, 0.5)$	—	0 ; 0	0/0 ; 0/0	—	—
Four-call cap $(0.5, 0.5)$	0.375	0 ; 128	0/0 ; 0/0	—	—

TABLE IV

LOCAL-MODEL VALIDATION WITH EXPLICIT CLAIM DENOMINATORS.

Quantity	Original v1	Revised v2
Completed/planned episodes	12/12	12/12
Actual calls/profile cap	36/36	23/36
Definite final status claims	12/12	0/12
Contradictions/assessable claims	6/10	0/0
State satisfied/violated/unknown	6/4/2	7/3/2
Linked satisfied/false/unknown	6/4/2	5/5/2

action sequences; this is not a pure reporting experiment or an attribution-method comparison.

The revised profile's conditional contradiction rate is undefined, not zero error. The original profile's 6/10 contradiction proportion has a descriptive 95% Wilson interval of approximately $[0.313, 0.832]$. Shared cases, fixed profile order and two repetitions per case preclude interpreting this interval as calibrated uncertainty over a broader population of tasks or models. In the other-actor case, revised-profile readback satisfied state while origin was external and action linkage false. In the expired-request case, the original profile correctly described ineligibility but issued a refund; its failure assessment contradicted achieved effect. Effect satisfaction and eligibility-policy correctness are distinct targets.

A separate unblinded annotation pass within the AI-assisted development workflow marked affirmative processing clauses in 10/12 original and 7/12 revised final rationales. Five of those seven revised rationales also hedged independent

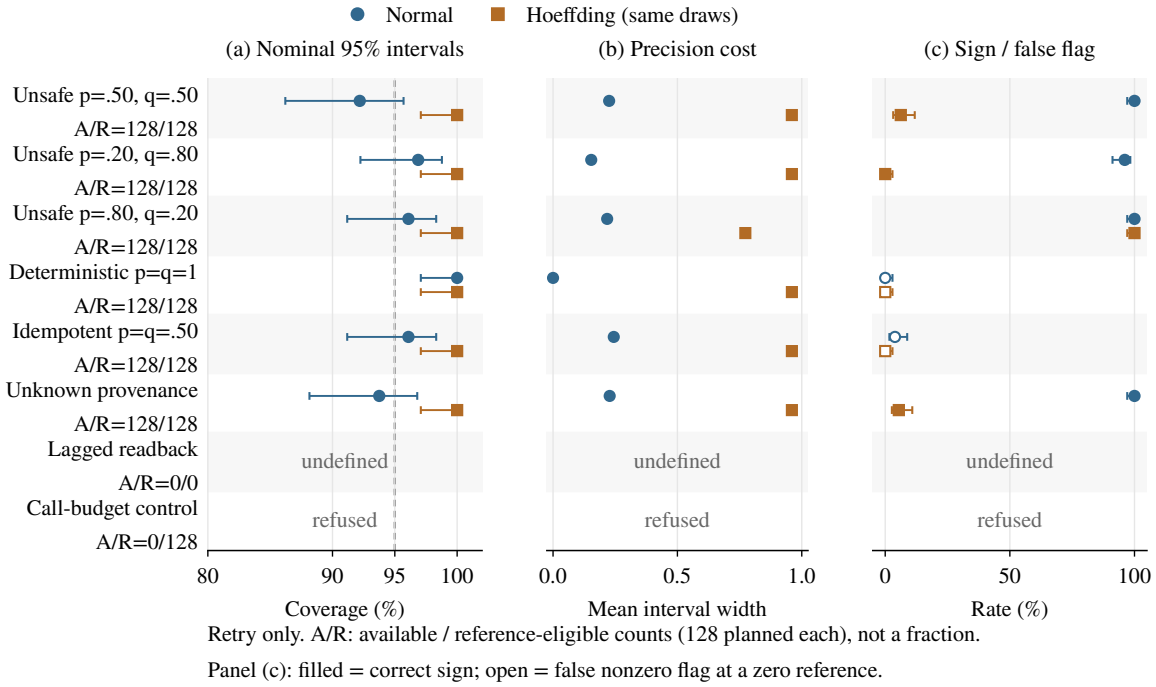


Fig. 4. On-reach global-consequence retry coefficient for all eight cases, using identical draws for both interval methods. A/R denotes available and population-reference-eligible counts, each from 128 planned trials. Coverage and flag whiskers are 95% Wilson uncertainty for repeated-trial rates, not coefficient intervals. Filled flag markers indicate correct directional detection; open markers indicate false nonzero flags for zero-reference controls. Undefined evidence and budget refusals have no zero-valued markers.

verification. A statement that a second attempt committed can be literally true while two effects violate an exactly-one task. These annotations are descriptive; no independent human verification or inter-rater reliability has been established. Unknown structured fields do not ensure prose abstention. The saved auditor rejected seven corrupted smoke copies without additional generation. Fixed profile order, tiny sample size, shared fault conditions and an exposed validation family limit interpretation. The held-out reserve was not generated.

VI. Discussion and Limitations

The evidence protocol supports different, inspectable claims about responses, state, origin and intervention effects. Its refusal rules prevent unsupported numerical outputs but can sharply reduce availability. Availability can depend on interventions and must remain visible beside numerical results. Conservative intervals do not make undefined games numeric, resolve missing evidence, or remove the need to assess the actual sampling law.

The environment is a narrow scripted simulator. No real payment API or standard world-state benchmark was evaluated. The artifact-delivery example validates an integration interface, not a second production-domain experiment. Declared source authority and watermarks are not authenticated transactional guarantees. The model experiments use one local model and exposed development/validation families; they do not estimate broad model performance or held-out transfer. State and linked targets sharing outcomes are not independent replications. Claims about hidden reasoning cannot be inferred from reported assessments.

Three evaluation gaps determine the next study design. First, compare a response-only auditor, a verify-before-retry wrapper, a faithful CAR-style adapter, an LLM trace judge and simple last-step rules on identical fault traces. State satisfaction, source-supported origin and causal intervention effects need separate reference labels. Report duplicate effects, false success claims, abstention/availability, attribution accuracy and actual cost under matched evidence, targets and call budgets. Joint context/fault attribution must expose infrastructure variables to every applicable method.

Second, assess bounded polling on an eventually consistent readback: poll until its declared watermark covers the horizon or a predeclared wait budget expires, then retain unknown. Report availability against wait time, read count and latency. Polling cannot promise finality or exactly-once execution under arbitrary late commits. The retained lag control uses one insufficient readback and yields zero consequence estimates; a polling result has not been measured. A safe real-service study would use provider test mode, isolated accounts and injected acknowledgement faults; counterfactual replay would remain in an isolated simulator or a separately validated digital twin.

Third, preregister held-out template and fault-composition families, multiple model classes and independent blinded double annotation with agreement reporting. A planned 3–5-model, 500-or-more-episode study requires a precision and clustering analysis before fixing its final sample size. At least

1,000 independent repetitions per selected simulator setting would better resolve coverage, with simultaneous uncertainty over predeclared positions and modern bounded-mean interval comparators. These are proposed experiments. No matched baseline, polling study, multi-model result or human annotation agreement is claimed here.

The retained cost record counts actual scripted calls and refusals; the repeated study used 628,535 calls and zero model calls. It does not measure per-call verification latency, trace-size overhead, real-API charges or cost per usable production estimate. Those measurements belong in the next evaluation.

VII. Conclusion and Artifact Availability

Consequa-X provides a reusable evidence core and auditable replay adapters for separating reported assessments, external outcomes, origin and intervention effects. The retained studies expose uncertainty failure, precision costs, variable-length game restrictions and prose/structured-status disagreement. The useful result is a reproducible protocol/software artifact with measured limits, without claiming a new estimator or general method superiority.

Stable version 1.1.0 is publicly available under MIT on PyPI [20]. Its wheel and source distribution were independently downloaded and hash-checked, and an isolated registry installation was verified on October 17, 2026. The wheel contains the runtime, schemas and typing marker; the source distribution also includes documentation, offline examples, tests and audit scripts. Private research outputs and model weights are excluded. The maintainer-controlled repository at https://github.com/Devrajsinh-Jhala/consequa-x is private. Git history, retained protocols, raw evidence and evaluator-only truth are retained there for authorized collaborators; this paper does not assert unrestricted access to that research archive. The experiments predate the 1.1.0 release: original calibration used source `9e2dcbb`, shared-draw comparison `0d8ec0b`, refund enumeration and local validation `2c4adfe`, and repeated sampling `0bd744d`. Full revisions and manifests are retained in the evidence ledger. Version 1.1.0 adds current engineering interfaces and offline reports; its publication does not rerun or change those historical studies. A public immutable research archive and review-access arrangements have not been established. Framework imports and offline examples require no model credentials or required runtime dependencies. All reported payments are simulated; model weights are identified rather than redistributed.

AI Assistance and Verification Disclosure

Generative AI assistance was used for software development, research-source inspection, rationale annotation and manuscript drafting. Recorded automated checks cover retained-evidence accounting, analytic references, numerical implementations and document compilation. They do not constitute independent human replication or blinded annotation. The rationale labels remain unblinded and exploratory. The author is responsible for the paper's claims and any venue-required disclosure; no independent human validation is asserted.

Appendix A
Declared-Game Properties and Test Evidence

Strict-game partiality. In the declared three-position mechanism, let $N = \{s, r, t\}$ be selection, timeout retry and final stop. If $p < 1$, the coalition $\{r\}$ requests a retry that is unreachable with probability $1 - p > 0$ when fresh selection declines. If $q < 1$, coalition $\{s, t\}$ holds selection and requests a final stop that is unreachable with probability $1-q > 0$ when the fresh timeout decision stops. With compatible decisions, normal execution, adequate budget and known outcomes, the full strict coalition game is structurally defined if and only if $p = q = 1$ in this mechanism. In particular the full strict game is partial at every interior (p, q) and ordinary full-game Shapley is undefined there. Conditioning on valid replay would define a different game. A finite sample with no invalid draw cannot establish zero population invalid probability.

On-reach allocation. Fix finite factual positions, an intervention mapping and a continuation law. Suppose each coalition terminates almost surely, all reached held decisions are compatible, and a bounded measurable numeric outcome exists almost surely. Ignoring holds beyond normal termination then defines $v(S) = \mathbb{E}[X_S]$ for every coalition. Its ordinary Shapley values obey efficiency, symmetry, the null-player property and linearity. To see efficiency, sum the marginal contributions along each permutation: they telescope to $v(N) - v(\varnothing)$, and averaging preserves the equality. Relabelling permutations gives symmetry; zero marginals give the null-player property; linearity follows from linearity of each difference and of averaging. These properties belong to this declared game. They establish neither causal identifiability nor nonnegative normalized responsibility. Unknown observations, incompatible holds and provider failures remain outside these assumptions. Finite independently sampled coalition values can give a nonzero estimate for a true null player, and an independently sampled endpoint need not equal the sum of finite estimated shares. Uniform forward permutations and their reverses each have a uniform marginal. With unbiased coalition samples, their paired average is unbiased for the declared Shapley value. Independent pairs, rather than individual walk steps, are the sampling units for the bounded-mean interval.

Test mapping. C1 is exercised by external-actor and absent-provenance tests in `test_verification.py` and conflicting-observation tests in `test_graph_identity.py`. C2 is exercised by `test_refund_interactions.py`, the exact-rational analytic checker, and retained enumeration controls. C3 is exercised by unknown-draw and exhausted-budget tests in `test_decision_shapley.py`, plus the saved-evidence branch audit. Tests establish behavior in their stated cases, not correctness of an untrusted source or a real-service policy.

Appendix B
Supporting Audit Inventory

Raw assertion counts are retained for reproducibility rather than treated as evidence strength: 3,283,523 for the shared-draw audit, 27,277 for the enumeration audit, 44,698,925 for repeated sampling, and 2,227 for the local validation. These counts depend on data volume and auditor implementation. More informative checks include independently recomputed summaries, exact-rational analytic references, source-bound inventories, refusal denominators and rejected semantic corruptions. None supplies independent human verification.

References

[1] J. Shah, "Causal Agent Replay: Counterfactual Attribution for LLM-Agent Failures," arXiv:2606.08275, v1, Jun. 2026.

[2] R. Meng, "Policy-Conditioned Counterfactual Credit for Verifiable Reinforcement Learning of Long-Horizon Language Agents," arXiv:2606.05263, v1, Jun. 2026.

[3] A. Sharma, "CausalLoss-Fin: Attributing Financial-Agent Loss to Decisions and Infrastructure Faults," arXiv:2609.25960, v1, Sep. 2026.

[4] I. K. Mansoor, A. Phadke, and P. Rana, "Verified Tool Calls Improve LLM Agent Reliability Under Non-Atomic Failures," arXiv:2608.02645, v1, Jul. 2026.

[5] Z. Wang, "Callability Is Not Operability: Controlled Interface Interventions for LLM Agents," arXiv:2608.23628, v2, Aug. 2026.

[6] S. Zhang *et al.*, "Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems," in *Proc. 42nd Int. Conf. Machine Learning*, PMLR, vol. 267, 2025, pp. 76583–76599. [Online]. Available: https://proceedings.mlr.press/v267/zhang25cq.html

[7] L. Moreau and P. Missier, Eds., "PROV-DM: The PROV Data Model," W3C Recommendation, Apr. 30, 2013. [Online]. Available: https://www.w3.org/TR/2013/REC-prov-dm-20130430/

[8] A. Trofimov and B. Novikov, "When Tool Calls Succeed but Workflows Fail: Anomalies at the Agent-Tool Boundary," arXiv:2609.15397v1, 2026. [Online]. Available: https://arxiv.org/html/2609.15397v1

[9] J. Li, "Where Does Exactly-Once Live? Model, Harness, and Tool-Contract Effects on Duplicate Side Effects in LLM Agents," arXiv:2609.29095v1, 2026. [Online]. Available: https://arxiv.org/html/2609.29095v1

[10] P. Helland, "Idempotence Is Not a Medical Condition," *Communications of the ACM*, vol. 55, no. 5, pp. 56–65, 2012, doi: 10.1145/2160718.2160734.

[11] W. Vogels, "Eventually Consistent," *Communications of the ACM*, vol. 52, no. 1, pp. 40–44, 2009, doi: 10.1145/1435417.1435432.

[12] H. Trivedi *et al.*, "AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents," in *Proc. 62nd Annu. Meeting Association for Computational Linguistics*, vol. 1, 2024, pp. 16022–16076, doi: 10.18653/v1/2024.acl-long.850.

[13] J. Lu *et al.*, "ToolSandbox: A Stateful, Conversational, Interactive Evaluation Benchmark for LLM Tool Use Capabilities," in *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025, pp. 1160–1183, doi: 10.18653/v1/2025.findings-naacl.65.

[14] S. Yao, N. Shinn, P. Razavi, and K. Narasimhan, "τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains," arXiv:2406.12045v1, 2024. [Online]. Available: https://arxiv.org/html/2406.12045v1

[15] L. D. Brown, T. T. Cai, and A. DasGupta, "Interval Estimation for a Binomial Proportion," *Statistical Science*, vol. 16, no. 2, pp. 101–133, 2001, doi: 10.1214/ss/1009213286.

[16] I. Waudby-Smith and A. Ramdas, "Estimating Means of Bounded Random Variables by Betting," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 86, no. 1, pp. 1–27, 2024, doi: 10.1093/jrsssb/qkad009.

[17] L. S. Shapley, "A Value for N-Person Games," RAND Corporation, P-295, 1952, doi: 10.7249/P0295.

[18] C. J. Clopper and E. S. Pearson, "The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934, doi: 10.1093/biomet/26.4.404.

[19] W. Hoeffding, "Probability Inequalities for Sums of Bounded Random Variables," Journal of the American Statistical Association, vol. 58, no. 301, pp. 13–30, 1963.

[20] D. Jhala, "Consequa-X," MIT Python distribution, ver. 1.1.0, Oct. 17, 2026. [Online]. Available: https://pypi.org/project/consequa-x/1.1.0/